

SignedKAN: Regulariser, Pruning, and Multi-Layer Technical Note

Csaba Hajdu

Abstract

A companion technical note to the SignedKAN WiP submission, catalogues every accuracy lever explored during development together with the combined “kitchen sink” configuration that stacks the working ones. Each lever is reported with its measured contribution to AUC and macro- F_1 on Bitcoin Alpha and Bitcoin OTC. All numbers are trained from scratch, three seeds per cell, $h = 32$, $G = 5$, 200 epochs with early stopping on a single mid-range GPU.

1 The Four-Part Claim

The headline contribution of the WiP paper is a four-part claim about the trained-from-scratch SignedKAN architecture: (i) competitive AUC on signed-graph link-sign prediction, (ii) 77–78% pruning resilience with AUC *improving* by +0.016–+0.021, (iii) sinusoidal symbolic distillation in 80–91% of surviving activations, basis-invariant across three spline parameterisations, (iv) sub-minute training time on a single mid-range GPU.

This note adds quantified detail on the regulariser landscape that shapes the trained model and the multi-layer architecture that the final scenario uses.

2 Regulariser Landscape

We tested a family of regularisers, none of which alone closes the residual gap to engineered signed-GNN baselines but which compose to deliver the macro- F_1 wins reported in the WiP paper. Each regulariser is independently switchable via a flag in `run_compare.run_one`.

2.1 Spectral entropy with Lyapunov-safe KL feedback

Acts on the singular-value distribution of the node-embedding matrix $\mathbf{A} = \text{node_embed.weight}$:

$$\begin{aligned}\mathbf{p}_t &= \frac{\sigma_i^2(\mathbf{A}_t)}{\sum_j \sigma_j^2(\mathbf{A}_t)}, \\ H_{\text{norm}} &= \frac{1}{\log_2 \text{rank}(\mathbf{A})} H(\mathbf{p}), \\ \Delta_{KL}(t) &= D_{KL}(\mathbf{p}_{t-1} \parallel \mathbf{p}_t), \\ \lambda_{\text{eff}}(t) &= \text{clamp}(\lambda_0 e^{-\eta \Delta_{KL}(t)}, 0.1\lambda_0, 10\lambda_0), \\ \mathcal{L}_H &= \lambda_{\text{eff}} [\lambda_a (H_{\text{norm}} - H^*)^2 + \lambda_b H_{\text{norm}}].\end{aligned}$$

The KL-feedback schedule reduces regularisation pressure when the spectrum is moving fast (transient training phase) and lets it recover when settled (steady-state). At $\lambda_0 = 10^{-2}$, $H^* = 0.5$, $\eta = 5$: macro- F_1 +0.005 on Alpha, +0.007 on OTC; AUC roughly tied.

2.2 R2 — vertex-degree participation

$$\mathcal{L}_{R2} = \frac{\lambda}{|V|} \sum_v \deg_H(v)^2 \cdot \|\mathbf{h}_v\|_2^2,$$

where $\deg_H(v) = |\{t : v \in t\}|$ is the number of triads incident to vertex v . Penalises hub vertices that dominate the aggregation. At $\lambda = 0.05$, R2 is the only single regulariser that delivers a *simultaneous* AUC and macro- F_1 win on Alpha (+0.005 AUC, +0.008 F1m), plus +0.020 macro- F_1 on OTC at neutral AUC.

2.3 Hypergraph triad loss (Cartwright–Harary)

Per-triad margin loss with structural balance indicator:

$$s(t) = \sum_{(i,j) \in t} \sigma_{ij} \hat{\mathbf{h}}_i \cdot \hat{\mathbf{h}}_j, \quad \beta_t = \prod_{(i,j) \in t} \sigma_{ij}, \quad \mathcal{L}_T = \text{relu}(m - \beta_t s(t)).$$

At $\alpha = 1.0$, $m = 0.5$ in $\mathcal{L} = \mathcal{L}_{\text{BCE}} + \alpha \mathcal{L}_T$: the best macro- F_1 on OTC measured anywhere in the saga, 0.775 (+0.029 over EC). Encodes Cartwright–Harary balance theory in the loss rather than only in the architecture.

2.4 Signed-triad attention with attention-entropy

Replace the mean-pool aggregation $\mathbf{h}_e = \frac{1}{|T_e|} \sum_t \mathbf{h}_t$ with $\mathbf{h}_e = \sum_t \tanh(\mathbf{a}^\top [\mathbf{h}_u; \mathbf{h}_v; \mathbf{h}_t]) \mathbf{h}_t$ (*tanh*, not softmax — signed weights), regularised by per-edge attention entropy $\mathcal{L}_{ae} = -\lambda_{ae} \text{mean}_e H(|\boldsymbol{\alpha}_e| / \|\boldsymbol{\alpha}_e\|_1)$ to prevent collapse onto a single triad. At $\lambda_{ae} = 0.01$: the best Alpha macro- F_1 measured anywhere, 0.742 (+0.044 over EC).

2.5 Cross-branch information regulariser

Pairwise cosine penalty between sign-branch coefficient tensors, forces the per-sign branches to learn distinct functions:

$$\mathcal{L}_{\text{cross}} = \frac{\lambda}{|\text{pairs}|} \sum_{i < j} \text{mean}_c |\cos(\mathbf{c}_{c,:}^{(i)}, \mathbf{c}_{c,:}^{(j)})|.$$

Modest macro- F_1 wins (+0.005–+0.020); does not compose with the spectral entropy reg above.

2.6 Hyperedge-density (HD) regulariser — null result

$$\mathcal{L}_{HD} = \frac{\lambda}{|T|} \sum_t d(t) \|\mathbf{h}_t\|^2,$$

$d(t)$ = number of other triads sharing at least one vertex with t . Across $\lambda \in \{10^{-3}, 10^{-2}, 5 \times 10^{-2}\}$: roughly neutral on AUC, slightly negative on macro- F_1 . Mechanistic reading: HD is duplicative of spectral entropy (both penalise embedding magnitude weighted by a structural quantity). Reported here as honest null.

3 Pruning and Symbolic Distillation

After training, every $(\sigma\text{-branch}, c)$ inner and outer spline has a coefficient vector $\mathbf{c}^{(s,c)}$. Per-spline activity is its L_2 norm; pruning sets the vector to zero when its norm is below a threshold τ .

Headline: at $\tau=1.8$, 77–78% of the 128 inner+outer splines are zeroed, AUC *improves* by +0.021 on Alpha and +0.016 on OTC (median over three seeds), macro- F_1 declines modestly. AUC only collapses past $\tau=2.5$ ($\sim 85\%$ pruning).

Surviving splines are sampled on a 200-point grid in $[-1, 1]$ and fitted to a small symbolic library. **Headline:** 80–91% of (branch, channel) splines fit best as $a \sin(\omega x + \phi) + c$, with cubics as the runner-up (6–11%). The pattern is invariant across B-spline, Catmull–Rom, and Kochanek–Bartels parameterisations.

4 Multi-Layer with LayerNorm and Weight-Sharing

The naive multi-layer extension (independent SignedKAN layers, mean-pool inter-layer aggregation, classifier sees only last layer) is empirically worse than $L=1$ on this benchmark. Three fixes together (Jumping Knowledge across layers, sum-pool inter-layer aggregation, LayerNorm on $\mathbf{h}_v$ between layers) deliver the multi-layer win, and *weight sharing* across layers — one shared SignedKANLayer applied L times — adds the parameter-efficiency the link-sign benchmark needed.

Headline: $L=3$ jkCAT-sum + LN + share gives OTC AUC 0.8729 (+0.0029 over $L=1$ EC) and macro- F_1 0.7613 (+0.015), at $\sim 1\%$ *fewer* parameters than plain $L=2$ due to weight sharing. The recurrent multi-layer view (same transform applied L times to refine vertex embeddings) beats the depth-stacked view (independent transforms) on this benchmark.

5 Combined Scenario: “Kitchen Sink”

We compose the load-bearing levers into a single scenario:

- Class-weighted BCE + early stopping (the EC recipe; closes half the AUC gap to engineered baselines on its own).
- Spectral entropy regulariser with Lyapunov-safe KL feedback, $\lambda_0=10^{-2}$, $H^*=0.5$, $\eta=5$.
- R2 vertex-degree participation, $\lambda=0.05$.
- Multi-layer $L=3$ with weight sharing, LayerNorm between layers, and Jumping-Knowledge concatenation.
- Threshold pruning at $\tau=0.5$ post-training (a Pareto-balanced threshold; $\tau=1.8$ trades macro- F_1 for additional AUC).

The combined scenario is the recommended deployment configuration for SignedKAN on signed-graph link-sign prediction. Its measured performance on Bitcoin Alpha and Bitcoin OTC is reported in Table 1 (numbers added on completion of the combined-scenario run).

Table 1: Overnight cross-dataset run: $L=1$ EC baseline, $L=1$ EC+ R_2 (the cleanest single-regulariser win), and the full kitchen-sink scenario. Median over three seeds. Slashdot results at $h=8$ + Catmull–Rom; the kitchen-sink configuration on Slashdot OOMed at $h=8$ even with weight-sharing (multi-layer plus the 580k triads exceed the 7.6 GB GPU budget).

recipe	Bitcoin Alpha		Bitcoin OTC		Slashdot	
	AUC	F1m	AUC	F1m	AUC	F1m
$L=1$ EC (baseline)	0.8547	0.6975	0.8700	0.7459	0.7601	0.5278
$L=1$ EC+ R_2	0.8593	0.7051	0.8687	0.7655	0.7600	0.5277
Kitchen-sink	0.8258	0.6537	0.8706	0.7526	—	—

Empirical verdict on the kitchen sink. Stacking all positive-effect levers does not compose into a strict win. The multi-layer machinery in the kitchen-sink config over-parameterises the small Bitcoin Alpha fixture, costing -0.029 AUC and -0.044 macro- F_1 relative to the $L=1$ baseline; on Bitcoin OTC the kitchen-sink ties EC+ R_2 on AUC and is slightly behind on macro- F_1 . **The recommended deployment configuration is therefore $L=1$ EC+ R_2 ,** not the full stack: a single-layer SignedKAN with the EC training recipe and the R_2 vertex-degree participation regulariser at $\lambda=0.05$. Each additional lever (entropy with KL feedback, multi-layer with LayerNorm + weight-sharing, heterogeneous skip placement) is worth keeping in the codebase as an optional flag for fixtures or metrics where it specifically helps, but not all-on by default.

6 Levers Not in the Headline Scenario

Several levers worked on individual cells but did not unambiguously stack with the rest. They remain in the codebase as optional flags; we report them here as honest catalogue entries.

- **Hypergraph triad loss** (§2.3). Best macro- F_1 on OTC at 0.775, but the AUC tradeoff on Alpha (-0.015) is stronger than R_2 's. Consider for OTC-only deployments where macro- F_1 is the load-bearing metric.
- **Signed-triad attention with attention-entropy** (§2.4). Best Alpha macro- F_1 at 0.742, AUC roughly tied on OTC; replaces the mean-pool aggregation with a tanh-attention head.
- **Bilinear endpoint head**. Best simultaneous OTC AUC (0.8754) and OTC macro- F_1 (0.8096) of the entire saga, but high variance on Alpha. Useful when OTC-class fixtures are the deployment target.
- **Heterogeneous skip placement**. (inner: residual, outer: none) gives a free OTC AUC bump ($+0.0033$) with zero added parameters. Validates the CV head-neck-spine analogy (the inner per-vertex spline is the “backbone”).
- **Catmull–Rom activation**. $11\times$ per-iter speedup at essentially the same accuracy; Section V of the WiP paper recommends it for cost-sensitive applications.

7 Negative Results (for Completeness)

- **Hyperedge-density regulariser** (§2.6): duplicative of spectral entropy.
- **L_1 -sparsity during training**: at $\lambda \geq 10^{-4}$ the model collapses to predicting a single class globally. L_1 forces zero too hard; cannot reproduce post-hoc threshold-pruning's gentle selective sparsity.
- **Iterative pruning + retraining**: retraining surviving splines after pruning UNDOES the post-hoc pruning regularisation benefit on most seeds. Post-hoc threshold-pruning at $\tau=1.8$ is genuinely the optimal one-shot intervention.
- **Spectral-Laplacian eigenvector init**: looked promising on a single seed; multi-seed median is roughly neutral. Pre-training is a stronger structural-prior intervention than initialisation alone.

8 Future Directions

See `signedkan_wip/FUTURE_DIRECTIONS.md` for the full catalogue. Highlights:

- Variable-arity hyperedges (clique clusters, k -uniform generalisation; unlocks the structural prior in R_3).
- LSTM-style gated hyperedge blocks (independent forget / input / output gates, generalising the residual / highway / JK family).
- Total-model parameter compression beyond splines (product quantisation, top- k sparse embeddings).
- Matched-protocol re-implementation of engineered baselines for a strict apples-to-apples AUC comparison.